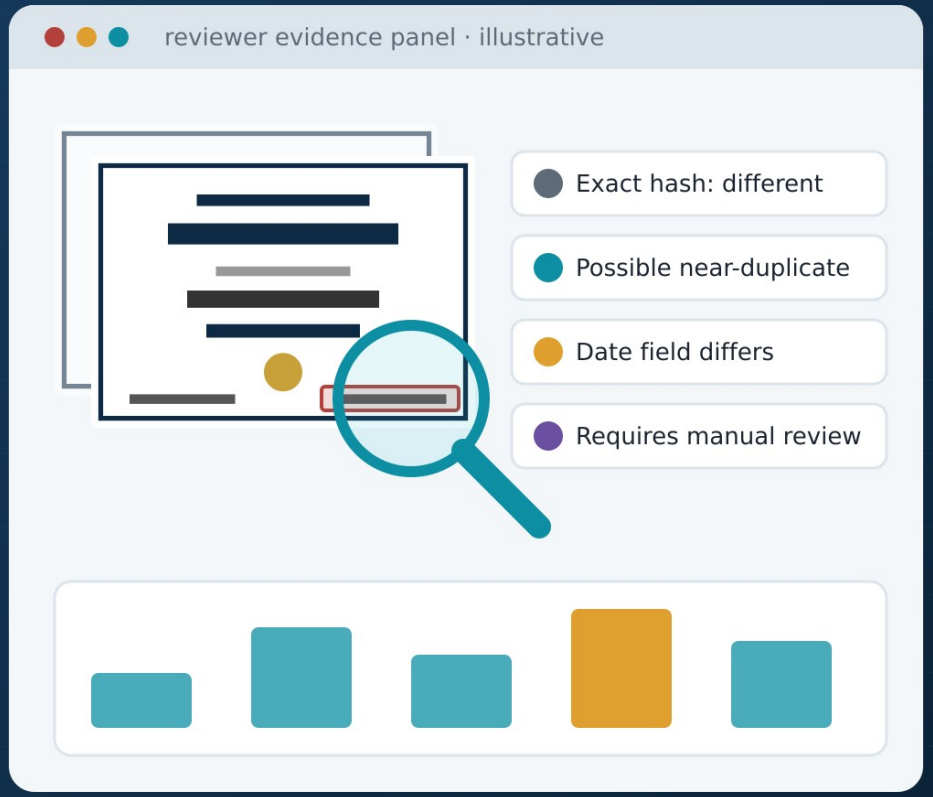
DISSERTATION PROPOSAL PRESENTATION

DocuGuard AI

An Explainable AI-Based Document Tampering and Duplicate Submission Detection Platform

Python · FastAPI · React + TypeScript · PostgreSQL · OCR · Computer Vision



Presented by	[Student Name] · [BITS ID]
Supervisor	[Supervisor Name], [Designation]
Organisation	[Organisation / Mentor Name]

BITS Pilani · CSIW ZG628T Dissertation · Academic Session 2026–2027

OVERVIEW

Presentation Outline

What we will cover in the next 10–15 minutes

01 Problem and motivation

02 Proposed solution

03 Objectives and scope

04 Architecture and workflow

05 Core detection methods

06 Database design

07 Evaluation plan

08 Timeline and conclusion

THE PROBLEM

Why Document Screening Is Difficult

Organisations receive large volumes of uploaded documents

- Repeated uploads of identical files
- Rescanned or format-converted copies
- Documents with modified dates or amounts
- Inconsistent text fields across versions
- Low-quality scans that are hard to read
- Slow, repetitive manual comparison

PROBLEM STATEMENT

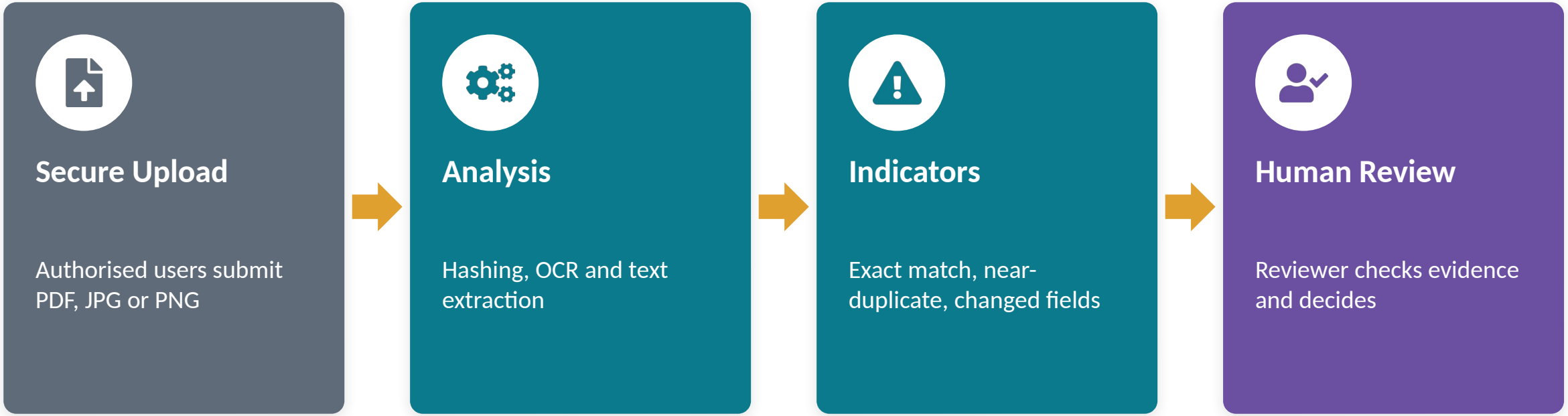
How can a lightweight platform identify exact and near-duplicate submissions and selected suspicious alterations while preserving reviewable evidence?

Why it matters
Simple file hashing misses rescans, and manual review does not scale — reviewers need clear, traceable evidence.

OUR PROPOSED SOLUTION

DocuGuard AI in One Picture

Automated indicators support — never replace — a human reviewer



In one sentence: DocuGuard AI combines hashing, OCR, text and region comparison to give reviewers explainable evidence of duplicates and selected changes — never a "fraud confirmed" verdict.

OBJECTIVES

What We Propose to Build

Initial focus: one or two fictional document templates (certificates, invoices)

Upload & registry

- Secure PDF/image upload
- Document metadata registry
- Restricted file storage

Analysis

- Exact duplicate detection
- Near-duplicate retrieval
- OCR / text extraction
- Field-difference analysis

Review & evaluation

- Evidence panel and review queue
- Reviewer decisions and audit trail
- Controlled experimental evaluation

RESEARCH CORE

an explainable combination of complementary screening signals inside a human-review workflow.

SCOPE

What Is In — and Out — of the First Release

A bounded scope keeps the 16-week dissertation feasible

In scope

- PDF, JPG and PNG files
- One or two fictional templates
- Exact duplicate detection
- Near-duplicate retrieval
- OCR where needed
- Selected field-level comparison
- Reviewer case management and evidence reports
- Basic audit logging

Out of scope / future work

- Issuer database verification
- Detecting every kind of forgery
- Legal authenticity decisions
- Automatic rejection
- Every document format
- Deep-learning training infrastructure

DocuGuard screens documents and supports reviewers — it cannot certify that a document is genuine.

SYSTEM ARCHITECTURE

How React, FastAPI, Python and PostgreSQL Work Together

Proposed layered design

User / Reviewer — Browser access, authenticated by role

↓

React + TypeScript — Upload screens, results, reviewer evidence panel

↓

FastAPI backend — Authentication, upload validation, APIs, job status

↓

Document processing — Hashing, PDF parsing, OCR, similarity and difference analysis

PostgreSQL

Metadata, analysis runs, matches, evidence, decisions and audit history

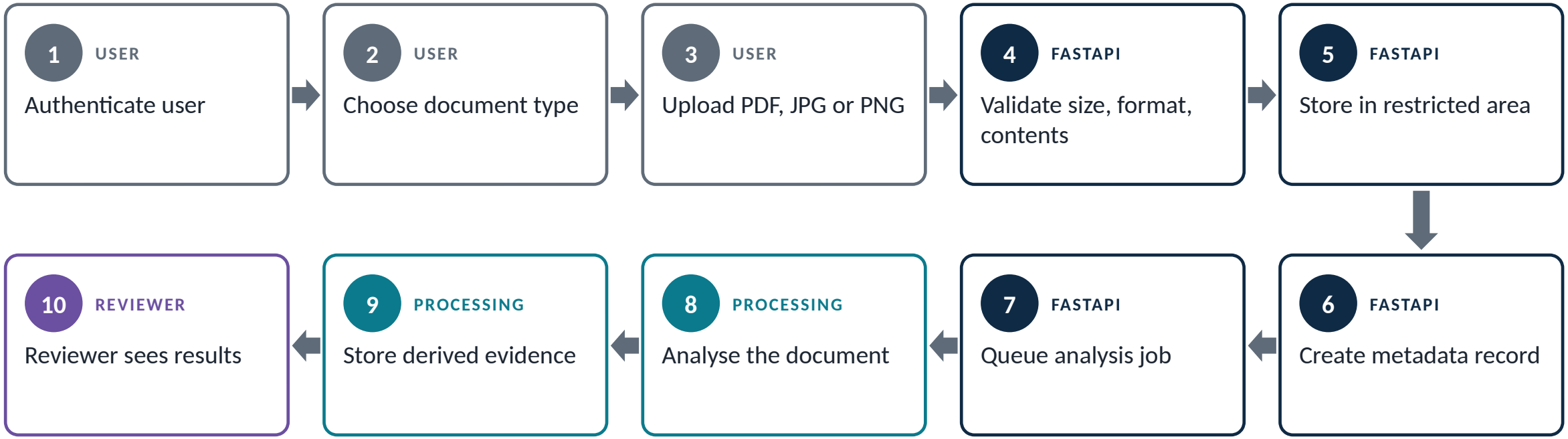
Restricted storage

Original files kept outside public folders; accessed only through the backend

END-TO-END WORKFLOW

What Happens When a Document Is Uploaded

Proposed processing lifecycle



Security: file extensions are never trusted alone — the server checks actual file contents before processing.

CORE METHOD 1

Exact and Near-Duplicate Detection

Two kinds of hashing for two different questions

HOW IT WORKS

- SHA-256 gives a fixed fingerprint of the file bytes
- Same digest → strong evidence of byte-identical files
- Perceptual hashing compares how pages look
- Lower hash distance suggests a near-duplicate candidate

Which method catches which case? (expected behaviour)

Case	SHA-256	Perceptual hash
Exact re-upload	Match	Match
Recompressed copy	Different	Close
Legitimate rescan	Different	Close
Small date edit	**Different**	**May still look close**
Unrelated document	Different	Far

Neither hash explains what changed — that needs text and field comparison.

CORE METHOD 2

OCR and Field-Level Comparison

Finding which specific values differ

HOW IT WORKS

- Text PDFs: extract text directly (PyMuPDF)
- Scanned images: OCR with Tesseract
- Normalise text, then compare key fields
- OCR errors can create false differences, so evidence is kept for review



Fictional certificate — issue date 12-06-2025 → 12-06-2026 highlighted

Indicator: "Issue date changed" — the reviewer verifies whether it is legitimate.

CORE METHOD 3

The Combined Explainable Pipeline

Complementary signals brought together for the reviewer

PIPELINE

- Validate and store → SHA-256 exact match
- Normalise page → perceptual hash
- Extract text / OCR → field comparison
- Region analysis → evidence report → reviewer

Exact hash: different
Not a byte-identical upload

Perceptual similarity: candidate near-duplicate
Page looks very similar to an earlier submission

Extracted text: one date field differs
Difference located near the issue-date area

System status: requires manual review
The highest automated status — never "fraud confirmed"

DATABASE DESIGN

What PostgreSQL Will Store

Conceptual entities — final schema designed during implementation

USER	DOCUMENT	DOCUMENT_FILE
role	submitted_by · type	sha256 · storage_ref

ANALYSIS_RUN	DOCUMENT_MATCH	DIFFERENCE_EVIDENCE
status · method versions	matched doc · type	field / region · description

REVIEW_CASE	REVIEW_DECISION	AUDIT_EVENT
run · state	reviewer · outcome	actor · action · time

KEY RELATIONSHIPS

- A user submits many documents
- Each document has a protected file reference
- An analysis run produces matches and evidence
- A review case references the analysis findings
- An authorised reviewer records the decision
- Audit events preserve important actions

EVALUATION PLAN

How We Will Evaluate the System

Planned methodology — no experiments have been run yet

WHAT WE WILL MEASURE

- Precision and recall
- F1-score
- False-positive rate
- Results per transformation type
- Evidence quality (correct region?)
- Processing time per document

ALSO TESTED

Upload validation, cross-user access denial, malformed files and reviewer decisions (Pytest).

Baseline comparison

RESULTS PENDING

Method	Exact dup.	Near dup.	Changes
SHA-256 only	TBD	TBD	TBD
OCR similarity only	TBD	TBD	TBD
Combined pipeline	TBD	TBD	TBD

We will build a controlled dataset of about 150–250 fictional documents with nine kinds of known changes. All variants of one source document stay in the same data split to avoid leakage.

PROJECT PLAN

Implementation Timeline and Milestones

Proposed 16-week work plan



2 Dec 2026
Mid-semester report

20 Jan 2027
Final report

Week 16
Final demonstration

CONCLUSION

Expected Outcomes and Limitations

What we expect to deliver — and what we will not claim

Expected outcomes

- A working, reviewer-centred screening web app
- Exact and near-duplicate detection with evidence
- Field-level change indicators for review
- A measured comparison against two baselines

Limitations

- Results apply to our fictional templates only
- OCR and scan quality affect accuracy
- Does not detect every type of forgery
- Never a legal or official authenticity decision

Upload securely, compare with complementary methods, present reviewable evidence, require human verification, evaluate with controlled experiments.

THANK YOU

Thank You — Questions?

We welcome feedback from our supervisor and reviewers

KEY TAKEAWAYS

1. SHA-256 finds exact duplicates; perceptual hashing finds look-alikes
2. OCR and field comparison show what changed
3. The reviewer always makes the final decision
4. Every claim will be tested on a controlled dataset

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DocuGuard AI is a proposed project. All examples, screens and values in this presentation are illustrative.